CS 3244

Machine Learning

Week 11 Post-Class
PCA Supplemental Notes on Reconstruction Loss

11 post

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PCA minimizing reconstruction loss

$$J_{recon} = m \sum_{k=K+1}^{n} \mathbf{v}_k \boldsymbol{\Sigma} \mathbf{v}_k^\top$$

$K+1$ to n as we are assessing loss, not fidelity

Let $\{u_1, \ldots, u_n\}$ be (row) eigenvectors of $\boldsymbol{\Sigma}$ and the corresponding eigenvalues be $\{\lambda_1, \ldots, \lambda_n\}$

Recall that:

$$\boldsymbol{\Sigma} = \sum_{i=1}^{n} \lambda_i u_i^\top u_i$$

J_{recon}

$$J_{recon} = m \sum_{k=K+1}^{n} \mathbf{v}_k \boldsymbol{\Sigma} \mathbf{v}_k^\top$$

$$= \sum_{i=1}^{n} \lambda_i \underbrace{\sum_{k=K+1}^{n} \mathbf{v}_k \, \mathbf{u}_i^\top \mathbf{u}_i \mathbf{v}_k^\top}_{\mathrm{w}_i}$$

$$= \sum_{i=1}^{n} \lambda_i \, w_i$$

Switch summation order

Observations:

- What if $w_i \in [0,1]$ and $\sum_{i=1}^{n} \mathrm{w}_i = n - k$?

  → w_i should be large when λ_i is large

  → The maximum value of J_{recon} is sum of largest k eigenvalues

$$w_i = \sum_{k=K+1}^{n} \mathbf{v}_k \, \mathbf{u}_i^\top \mathbf{u}_i \mathbf{v}_k^\top$$

Each term $\mathbf{v}_k \, \mathbf{u}_i^\top \mathbf{u}_i \mathbf{v}_k^\top = (\mathbf{u}_i \mathbf{v}_k^\top)^2 \geq 0$

$\rightarrow w_i \geq 0$

Since each term is positive :

$$w_i = \sum_{k=K+1}^{n} \mathbf{v}_k \, \mathbf{u}_i^\top \mathbf{u}_i \mathbf{v}_k^\top = \sum_{k=K+1}^{n} \mathbf{u}_i \, \mathbf{v}_k^\top \mathbf{v}_k \mathbf{u}_i^\top$$

$$w_i \leq \sum_{k=1}^{n} \mathbf{u}_i \, \mathbf{v}_k^\top \mathbf{v}_k \mathbf{u}_i^\top = \mathbf{u}_i \, \mathbf{V}^\top \mathbf{V} \mathbf{u}_i^\top = \mathbf{u}_i \mathbf{u}_i^\top = 1$$

$\rightarrow w_i \leq 1$

$$\sum_{i=1}^{n} \mathrm{w}_i = \sum_{i=1}^{n} \sum_{k=K+1}^{n} \mathbf{v}_k \, \mathbf{u}_i^\top \mathbf{u}_i \mathbf{v}_k^\top$$

$$= \sum_{k=K+1}^{n} \sum_{i=1}^{n} \mathbf{v}_k \, \mathbf{u}_i^\top \mathbf{u}_i \mathbf{v}_k^\top$$

$$= \sum_{k=K+1}^{n} \mathbf{v}_k \, \mathbf{U}^\top \mathbf{U} \mathbf{v}_k^\top$$

$$= \sum_{k=K+1}^{n} \|\mathbf{v}_k\|^2 = n - K$$

$\rightarrow \sum_{i=1}^{n} \mathrm{w}_i = n - k$